

Bank Customer Churn Analysis Report

Project Overview:

Customer churn is a critical challenge in the banking industry, as it directly impacts revenue, customer lifetime value, and overall business growth. This project focuses on analyzing customer data to understand the key factors influencing churn and to identify high-risk customer segments.

The dataset consists of 10,000 customer records, including demographic details (such as age, gender, and geography), financial attributes (such as balance and credit score), and behavioral indicators (such as account activity and complaints). The target variable, *Exited*, indicates whether a customer has churned or not.

This analysis is conducted using Python (Pandas, NumPy, Seaborn, Matplotlib) for data cleaning, feature engineering, and exploratory data analysis (EDA), along with SQL for extracting key business insights and performance metrics.

The project aims to:

- Understand customer behavior patterns
- Identify major drivers of churn
- Segment customers based on risk levels
- Provide actionable business recommendations to improve retention

By combining analytical techniques with business insights, this project helps in developing a data-driven approach to reduce customer churn and enhance customer satisfaction.

1. Python Analysis (EDA)

◆ 1.2 Dataset Overview

- Total Records: **10,000**
- Total Features: **18**
- Target Variable: **Exited (0 = Stayed, 1 = Churned)**

Dataset includes:

- Demographics → Age, Gender, Geography

- Financial → Balance, Credit Score
- Behavioral → Activity, Complaints

	RowNumber	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts
0	1	15634602	Hargrave	619	France	Female	42	2	0.00	1
1	2	15647311	Hill	608	Spain	Female	41	1	83807.86	1
2	3	15619304	Onio	502	France	Female	42	8	159660.80	3
3	4	15701354	Boni	699	France	Female	39	1	0.00	2
4	5	15737888	Mitchell	850	Spain	Female	43	2	125510.82	1
5	6	15574012	Chu	645	Spain	Male	44	8	113755.78	2
6	7	15592531	Bartlett	822	France	Male	50	7	0.00	2
7	8	15656148	Obinna	376	Germany	Female	29	4	115046.74	4
8	9	15792365	He	501	France	Male	44	4	142051.07	2
9	10	15592389	H?	684	France	Male	27	2	134603.88	1

NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited	Complain	Satisfaction Score	Card Type	Point Earned
1	1	1	101348.88	1	1	2	DIAMOND	464
1	0	1	112542.58	0	1	3	DIAMOND	456
3	1	0	113931.57	1	1	3	DIAMOND	377
2	0	0	93826.63	0	0	5	GOLD	350
1	1	1	79084.10	0	0	5	GOLD	425
2	1	0	149756.71	1	1	5	DIAMOND	484
2	1	1	10062.80	0	0	2	SILVER	206
4	1	0	119346.88	1	1	2	DIAMOND	282
2	0	1	74940.50	0	0	3	GOLD	251
1	1	1	71725.73	0	0	3	GOLD	342

◆ 1.3 Data Understanding

- No duplicate records found
- No missing values present
- Data types include:
 - Numerical → CreditScore, Age, Balance
 - Categorical → Gender, Geography, Card Type

Dataset is clean and ready for analysis

◆ 1.4 Data Cleaning

- Removed unnecessary columns:

- RowNumber
- CustomerId
- Surname

These do not contribute to churn prediction

◆ 1.5 Feature Engineering

To improve analysis, new features were created:

✓ Age Category

- Young customers
- Early professionals
- Mid-career
- Mature customers
- Pre-retirement
- Senior citizens

✓ Balance Group

- 0–50k
- 50k–100k
- 100k–150k
- 150k–200k
- 200k+

✓ Tenure Group

- 0–2 yrs
- 3–5 yrs
- 6–8 yrs
- 9–10 yrs

1.6 Exploratory Data Analysis (EDA)

◆ 1.6.1 Univariate Analysis

Categorical Variables

- France has the highest number of customers (~50%)
- Germany and Spain contribute ~25% each
- Gender distribution is nearly balanced
- Card types are evenly distributed

Numerical Variables

- Average Credit Score: ~650
- Average Age: ~39 years
- Balance shows wide variation with some outliers

No extreme anomalies detected

◆ **1.6.2 Bivariate Analysis**

Churn vs Complaints

- Customers with complaints → ~99.5% churn
Extremely strong relationship (possible data leakage)

Churn vs Geography

- Germany → highest churn (~32%)
- France & Spain → ~16%

Churn vs Age

- Middle-aged customers show higher churn
- Older customers (60+) show better retention

Churn vs Gender

- Female customers churn more than male customers

Churn vs Tenure

- Slight fluctuations observed
- No strong linear relationship

Churn vs Balance

- Higher balance customers tend to churn more

Churn vs Activity

- Inactive customers have significantly higher churn

Churn vs Credit Card

- No major difference observed

Churn vs Satisfaction Score

- Churn distribution is almost uniform

1.6.3 Multivariate Analysis

Geography + Gender + Churn

- Germany + Female → highest churn segment

Age + Geography + Churn

- Mature customers in Germany show very high churn

Activity + Tenure + Churn

- Inactive users consistently show higher churn across all tenure groups

Balance + Churn

- High-value customers show higher churn probability

1.7 Correlation Analysis

- Complain → **Very high correlation (~0.99)**
- Age → moderate correlation
- Activity → negative correlation

Indicates:

- Complaints dominate churn behavior
- Some variables have weak influence

1.8 Statistical Testing (Chi-Square)

Significant Factors:

- Geography
- Gender

- IsActiveMember
- Complain

Not Significant:

- HasCrCard

◆ 1.9 Python EDA Summary

Key findings from EDA:

- Complaints strongly influence churn (with leakage concern)
- Germany is the highest churn region
- Female customers show higher churn
- Inactive users are more likely to churn
- High-value customers are at risk
- Middle-aged customers have highest churn

2. Business Insights using SQL:

- **Overall Churn Rate** - Calculates the percentage of customers who have churned out of the total customers.

	churn_rate
▶	20.38

- **Highest Customer Geography**- Identifies the region with the highest number of customers

	Geography	total_customers
▶	France	5014

- **Average Credit Score** - Calculates the average credit score of all customers.

	avg_creditScore
▶	650.53

- **Complaint Percentage-** Calculates the percentage of customers who have raised complaints.

	complain_rate
▶	20.44

- **Churn Rate by Tenure Group** -Calculates the churn rate for different customer tenure groups.

Tenure_Group	ChurnRate_by_Tenure
0-2 yrs	21.15
6-8 yrs	18.87
3-5 yrs	20.76
9-10 yrs	21.37

- **Avg Credit Score by Age Group** - Shows how the average credit score varies across different age categories.

Age_category	avg_creditscore_agegroup
Mid-career	649.07
Early professionals	651.65
Mature customers	646.36
Young customers	651.14
Pre-retirement	647.99
Senior-citizen	667.32

- **Avg Balance by Tenure-** Analyzes how average account balance changes across tenure groups.

Tenure_Group	avg_balance_tenure
0-2 yrs	78053.98
6-8 yrs	75993.61
3-5 yrs	76070.91
9-10 yrs	75686.58

- **Total Churn by Tenure-** Counts the number of churned customers in each tenure group.

Tenure_Group	total_churn
0-2 yrs	528
6-8 yrs	570
3-5 yrs	625
9-10 yrs	315

- **Churn by Gender & Geography** - Compares churn rates across different gender and geographic segment.

Gender	Geography	churn_by_Gender_Geography
Female	France	20.34
Female	Spain	21.21
Male	Spain	13.11
Male	France	12.75
Female	Germany	37.55
Male	Germany	27.81

- **Avg Points by Card Type** - Identifies which card types have higher average reward points earned.

Card Type	avg_point_by_cardtype
PLATINUM	608.84
GOLD	607.24
DIAMOND	605.98
SILVER	604.00

- **Customer Segmentation (High vs Low Value)** -Segments customers into high-value and low-value groups based on balance and activity status.

Segment	total_customers
Low-Value	5201
High-Value Inactive	2356
High-Value Active	2443

- **Churn by Satisfaction Score** -Shows how churn is distributed across different satisfaction score levels.

Satisfaction Score	churn_by_SatisfactionScore
2	439
3	401
5	397
4	414
1	387

- **Complaint Impact on Churn** -Analyzes how customer complaints affect churn rate.

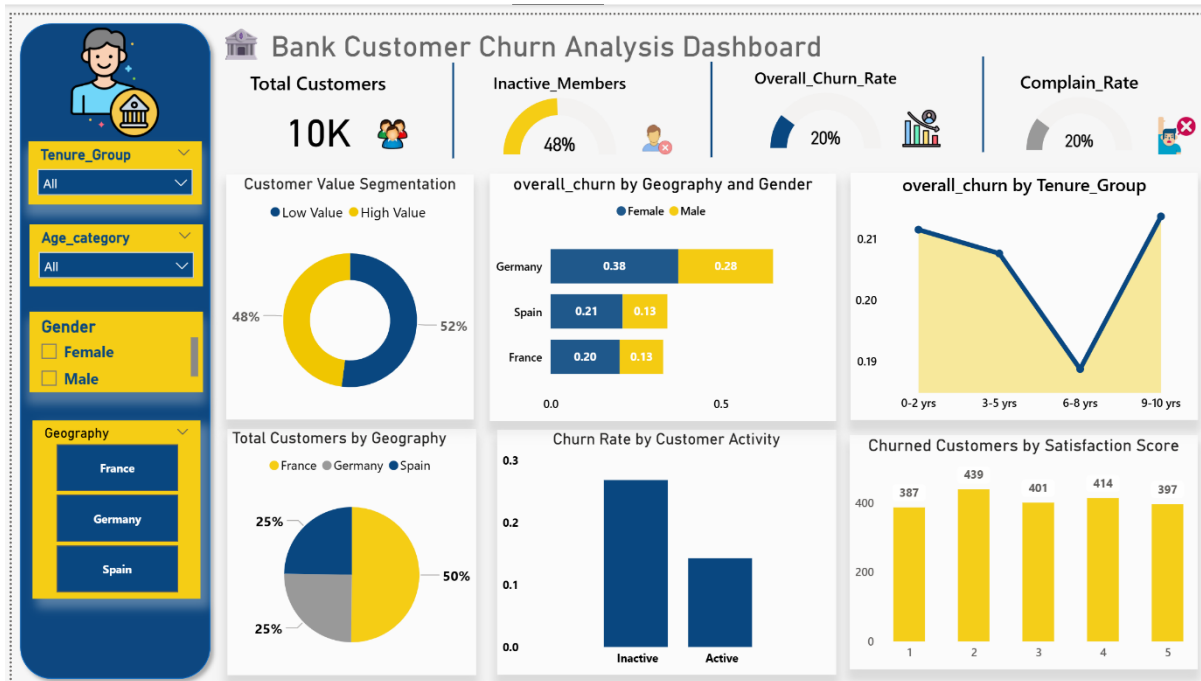
	Complain	ChurnRate
▶	1	99.51
	0	0.05

- **Churn by Age Group** - Calculates churn rates across different age categories.

Age_category	ChurnRate
Mid-career	30.83
Early professionals	10.88
Mature customers	56.04
Young customers	7.56
Pre-retirement	35.20
Senior-citizen	9.93

3. Dashboard Analysis & Interpretation

The dashboard provides a comprehensive overview of customer churn behavior using key performance indicators (KPIs), segmentation visuals, and comparative analysis across multiple dimensions.



3.1 Key Performance Indicators (KPIs)

- Total Customers: 10,000
- Overall Churn Rate: 20%
- Complaint Rate: 20%
- Inactive Members: 48%

Interpretation:

A churn rate of 20% indicates that a significant portion of customers is leaving. Additionally, nearly half of the customers are inactive, which aligns with higher churn risk observed in earlier analysis.

3.2 Customer Value Segmentation

- Low Value Customers → 52%
- High Value Customers → 48%

Interpretation:

The customer base is almost evenly split, but a substantial portion of customers falls into the high-value category, making retention strategies for these customers critical for business profitability.

3.3 Churn by Geography and Gender

- Germany shows the highest churn rates
- Female customers have higher churn than male customers across all regions

Interpretation:

Germany emerges as a high-risk market, and gender-based differences indicate the need for targeted retention strategies, especially for female customers.

3.4 Churn by Tenure Group

- Highest churn → 9–10 years
- Lowest churn → 6–8 years

Interpretation:

Churn follows a U-shaped trend, where new and long-term customers are more likely to leave, while mid-tenure customers show better retention.

3.5 Customer Distribution by Geography

- France → 50%
- Germany → 25%
- Spain → 25%

Interpretation:

France dominates the customer base, while Germany and Spain contribute equally. Despite fewer customers, Germany shows disproportionately higher churn.

3.6 Churn by Customer Activity

- Inactive customers → significantly higher churn (~27%)
- Active customers → lower churn (~14%)

Interpretation:

Customer engagement plays a crucial role in retention. Inactive users are much more likely to churn, making engagement strategies essential.

3.7 Churn by Satisfaction Score

- Churn is relatively evenly distributed across all satisfaction levels

Interpretation:

Satisfaction score alone does not strongly influence churn, indicating that other behavioral or external factors may play a bigger role.

4. Business Recommendations

- **Improve complaint resolution system to reduce churn risk**
- **Focus on high-churn region (Germany) with targeted strategies**
- **Provide personalized services for female customers**
- **Increase engagement to convert inactive users into active ones**
- **Retain high-value customers through premium benefits**
- **Strengthen retention for new and long-term customers**
- **Target middle-aged customers (46–65) with customized offerings**